Koneru Lakshmaiah Education Foundation
(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)
Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.
Phone No: 7815926816, www.klh.edu.in

OPERATING SYSTEMS AND SYSTEMS PROGRAMMING (25CS2104E)

2026–27, Term-I

Project Problem Statement Submission Form

Section / Team Details

Section No.: 6

Team No.: 2

Project Title: Argument Passing

Team Members

Roll Number	Student Name	Signature
2520030156	Y.Ratna priya	
2520030086	J.Hasini	
2520030132	N.Teja Sri	

1. Abstract

Argument Passing

Argument Passing is a system programming project that demonstrates how arguments can be passed from a parent process to a child process in a Linux environment. The project focuses on understanding how processes communicate and how command-line arguments are handled during program execution. It uses C programming and Linux process management mechanisms to create a simple and practical demonstration of argument passing.

The system accepts arguments from the command line and passes or makes them available to the required process for execution. It demonstrates the creation of processes, handling of process arguments, and execution of programs using Linux-based system programming concepts. The project helps in understanding process creation, command-line argument handling, and basic inter-process communication concepts.

The proposed solution will be implemented on Linux/Ubuntu using C programming and relevant POSIX/Linux APIs. The expected outcome is a working application that clearly demonstrates how arguments are received, processed, and passed between processes, providing a practical understanding of operating system and system programming concepts.

2. Problem Statement

Argument Passing project aims to demonstrate how arguments are passed between processes in a Linux environment. It focuses on process creation, argument handling, and execution using C programming and Linux system calls

3. Objectives

1. To understand the concept of command-line argument passing in Linux.
2. To demonstrate how arguments are received and processed by a C program.
3. To implement argument passing between processes using Linux system programming mechanisms.
4. To provide a simple practical demonstration of process creation and argument handling.

4.Proposed Methodology

The project will be implemented in a Linux/Ubuntu environment using C programming. The application will accept arguments through the command line and process the received arguments. A parent process can create a child process using appropriate Linux process management functions, and the required arguments can be provided to the child process during program execution.

The implementation will demonstrate the flow of arguments from input to processing and execution. Linux/POSIX system calls and process-related mechanisms will be used wherever required. The output will display the received and processed arguments so that the argument-passing mechanism can be clearly understood.

5. Operating Systems Concepts / Linux APIs Used

OS Concept / Linux API / System Call	Purpose in the Project
Process Creation	To create and manage processes
Command-Line Arguments	To receive input arguments during program execution
fork()	To create a child process
exec() family	To execute a program with specified arguments
Process Management	To demonstrate parent and child process execution
POSIX APIs	To implement Linux-based system programming functionality

6. Individual Contribution

(Clearly specify the responsibility of each team member in the project.)

Roll Number	Student Name	Individual Responsibility
2520030156	Y.Ratna priya	Project design and argument-passing implementation and Documentation and GitHub management
2520030086	J.Hasini	Process creation and Linux system calls

2520030132	N.Teja Sri	Testing and output verification

7. Tools / Platforms / Software Used

(List the programming languages, operating system, libraries, tools, and software planned for the project.)

Tool / Platform / Software	Purpose
Linux / Ubuntu	Development and execution environment
C / C++	Project implementation
GCC Complier	Compilation of C programs
POSIX API'S	Process and system programming operations
GitHub	Source code management and project repository

8. Expected Outcome

Expected Outcome

The project is expected to provide a working Linux-based application that demonstrates argument passing during program execution. The system will successfully accept command-line arguments, process them, and pass the required arguments to the appropriate process or program.

The project will provide a clear practical understanding of process creation, command-line arguments, process execution, and Linux system programming mechanisms. The final output will demonstrate the arguments received and processed by the program.

9. GitHub Repository

GitHub Repository Link: https://github.com/yratnapriya05/2520030156_86_132_OS_PROJECT

Repository Created: YES

□ Yes □ No

Faculty Name and Signature

Faculty Name:

Signature:

Remarks:

Faculty Use

Project Approved:

☐ Yes ☐ No

Date:

Faculty Signature: _____